

Question				Marking details				Marks available					
								AO1	AO2	AO3	Total	Maths	Prac
3	(a)							4			4		
				Property of the ball	Stays constant	Gets smaller	Gets larger						
				acceleration	✓								
				speed		✓							
				kinetic energy		✓							
				potential energy			✓						
				total energy	✓								
				1 mark for each correct row									
	(b)	(i)	Substitution: 0.46×10 (1) = 4.6 [N] (1)	1	1		2	2					
		(ii)	Substitution: 4.6 (ecf) $\times 15$ (1) = 69 (1) Joules or J or N m (1)	1 1	1		3	2					
		(iii)	[Yes because] acceleration [due to gravity] is the same as gravitational field strength or no other forces acting or $a = g$ or weight = 0.46×10 also or weight is 4.6 [N] or gravitational field strength = 10 or g is 10 Accept they're both 10 or we only multiplied by 10			1	1						
			Question 3 total	7	2	1	10	4	0				